Linux Directory Structure Notes

Linux organizes everything under the **root directory** `/`. Below are the most important system directories, their purpose, and equivalent locations in Windows and Mac.

1️⃣ `/` – Root Directory

- **What it is:** The "top folder" where everything starts.
- **Windows/Mac equivalent:** `C:\` on Windows, `/` on Mac.
- **What you do here:** Usually nothing directly; all other directories are inside this.
- **Example contents:** `/bin`, `/home`, `/usr`, `/etc`

2️⃣ `/bin` – Essential Commands

- **What it is:** Holds essential user commands.
- **Examples:** `ls`, `cp`, `mv`, `cat`
- **Windows/Mac equivalent:** `C:\Windows\System32`, `/usr/bin`
- **Layman:** Basic tools you always need.

3️⃣ `/sbin` – Admin/System Commands

- **What it is:** Commands for system administration, usually requiring root.
- **Examples:** `reboot`, `ifconfig`, `iptables`
- **Windows/Mac equivalent:** Windows admin tools, `/usr/sbin`
- **Layman:** Tools for IT admins, not regular users.

4️⃣ `/lib` – System Libraries

- **What it is:** Libraries (shared code) used by programs in `/bin` and `/sbin`.
- **Windows/Mac equivalent:** Windows DLLs (`C:\Windows\System32`), `/usr/lib`
- **Layman:** Ingredients needed for system programs to run.

5️⃣ `/boot` – Boot Files

- **What it is:** Contains files needed to boot Linux.
- **Examples:** `vmlinuz` (kernel), `initrd.img`, `grub/`
- **Windows/Mac equivalent:** `C:\Windows\Boot`, Mac boot partition
- **Layman:** Don't touch unless you know about booting Linux.

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6 `/etc` – Configuration Files

- **What it is:** System-wide configuration settings.
- **Examples:**
- `/etc/passwd` → user accounts
- `/etc/hostname` → system hostname
- `/etc/nginx/nginx.conf` → Nginx config
- **Windows/Mac equivalent:** Windows Registry/config files, `/Library/Preferences`
- **Layman:** The "settings folder" for your system and apps.

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7 `/dev` – Device Files

- **What it is:** Represents devices as files.
- **Examples:**
- `/dev/sda` → hard disk
- `/dev/tty` → terminal
- **Windows/Mac equivalent:** Device Manager (Windows), `/dev` (Mac)
- **Layman:** Your hardware appears as files here.

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8 `/proc` – Process & System Info

- **What it is:** Virtual directory providing info about processes and system.
- **Examples:**
- `/proc/cpuinfo` → CPU details
- `/proc/meminfo` → memory usage
- **Windows/Mac equivalent:** Task Manager (Windows), Activity Monitor (Mac)
- **Layman:** "System info files."

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9 `/tmp` – Temporary Files

- **What it is:** Stores temporary files created by applications.
- **Windows/Mac equivalent:** `C:\Windows\Temp`, `/tmp`
- **Layman:** Safe to delete; cleared on reboot.

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10 `/var` – Variable Data

- **What it is:** Data that changes frequently.
- **Examples:**
- `/var/log` → logs (`syslog`, `auth.log`)
- `/var/spool` → queues (mail, print jobs)
- `/var/tmp` → temporary files (survive reboot)
- **Windows/Mac equivalent:** Event Viewer logs, temp folders
- **Layman:** "Dynamic data folder."

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11 `/usr` – User Programs & Data

- **What it is:** Contains user applications, libraries, documentation.
- **Subfolders:**
- `/usr/bin` → apps (python3, gcc)
- `/usr/sbin` → admin apps (apache2)
- `/usr/lib` → libraries
- `/usr/local` → manually installed apps
- **Windows/Mac equivalent:** `C:\Program Files`, `/Applications`
- **Layman:** Where installed programs live.

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12 `/home` – User Home Directories

- **What it is:** Each user's personal space.
- **Example:** `/home/prashant`, `/home/user1`
- **Windows/Mac equivalent:** `C:\Users\username`, `/Users/username`
- **Layman:** Your "My Documents" on Linux.

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13 `/media` & `/mnt` – Mount Points

- **What it is:** For removable and temporary mounted filesystems.
- **Examples:** `/media/usb`, `/mnt/disk`
- **Windows/Mac equivalent:** `D:\`, `E:\` drives, `/Volumes/USB`
- **Layman:** Where USB drives, CDs, or external disks show up.

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14 `/opt` – Optional Software

- **What it is:** Stores optional or third-party applications.
- **Examples:** `/opt/google/chrome`, `/opt/eclipse`
- **Windows/Mac equivalent:** Custom apps in `C:\Program Files`, `/Applications`
- **Layman:** "Extra apps folder."

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15 `/run` – Runtime Data

- **What it is:** Stores runtime information such as process IDs (PID files), sockets, and lock files.
- **Layman:** Programs leave their "currently running" status here.

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16 `/srv` – Service Data

- **What it is:** Contains service-related files.
- **Examples:** `/srv/www` → website data

- **Windows/Mac equivalent:** `C:\inetpub\wwwroot`, `/srv`
- **Layman:** Only used by servers (web, FTP, etc.).

Summary Table

Directory	Purpose	Windows Equivalent	Mac Equivalent
`/`	Root directory	`C:\`	`/`
`/bin`	Essential commands	`C:\Windows\System32`	`/usr/bin`
`/sbin`	Admin commands	Admin tools	`/usr/sbin`
`/lib`	System libraries	DLL files	`/usr/lib`
`/boot`	Boot files	`C:\Windows\Boot`	Boot partition
`/etc`	Configuration files	Registry/config files	`/Library/Preferences`
`/dev`	Device files	Device Manager	`/dev`
`/proc`	Process/system info	Task Manager	Activity Monitor
`/tmp`	Temporary files	`C:\Windows\Temp`	`/tmp`
`/var`	Variable data (logs, queues)	Event Viewer/Temp	`/var`
`/usr`	User programs, libraries	`C:\Program Files`	`/Applications`
`/home`	User personal files	`C:\Users\username`	`/Users/username`
`/media`	Removable drives	`D:\`, `E:\`	`/Volumes/USB`
`/mnt`	Temporary mount points	`D:\`, `E:\`	`/Volumes/USB`
`/opt`	Optional software	`C:\Program Files`	`/Applications`
`/run`	Runtime data	N/A	`/var/run`
`/srv`	Service data	`C:\inetpub\wwwroot`	`/srv`

✅This breakdown shows how Linux directories map to equivalents in Windows and Mac, with practical examples.